Spectral Scale Selection in Topological Data Analysis

Ilja Sukhanov[1] Djordje Baralic[2] Oleksandr Leonov[3] Liudmyla Polyakova[3] Evgeny Spodarev[1]

[1] Universität Ulm, Germany [2] MISANU, Belgrade [3] Kharazin National University, Charkiw

1. Why Select One Scale

- Persistent homology is scale-agnostic: persistence diagrams record features **across all filtration scales**.
- Nevertheless, many problems in and around TDA reduce to the **choice of one filtration parameter**.
- This choice should be made in a **theoretically justified** way.

Reconstruction and meshing from point clouds | **Topology-aware clustering and graph construction**

Statistical inference for point processes | **Materials and physical systems with intrinsic length scales**

2. Spectral Scale Selection

- Let $K_1, \dots, K_N$ be simplicial complexes on a common vertex set X and $\oplus$ a commutative binary operation; below, $\oplus = \cup$.
- f combines topological or combinatorial features and is **subadditive** with respect to $\oplus$.

Motivated by **[1]**, define the f**-based divergence** and the **scale similarity matrix**:

$$\langle K_i, K_j\rangle_f \coloneqq f(K_i) + f(K_j) - f(K_i \oplus K_j), \qquad \Sigma_N \coloneqq \big(\langle K_i, K_j\rangle_f\big)_{i,j=1}^{N} \in \mathbb{R}^{N\times N}.$$

For a filtration $K_i = \mathrm{VR}_{\epsilon_i}(X)$ with $0 < \epsilon_1 < \cdots < \epsilon_N$,

$$\langle K_i, K_j\rangle_f = f(K_{\min\{i,j\}}),$$

so Σ_N is a classical **min/meet-matrix**:

$$\Sigma_N = \begin{pmatrix} f(K_1) & f(K_1) & \cdots & f(K_1) \\ f(K_1) & f(K_2) & \cdots & f(K_2) \\ \vdots & \vdots & \ddots & \vdots \\ f(K_1) & f(K_2) & \cdots & f(K_N) \end{pmatrix}.$$

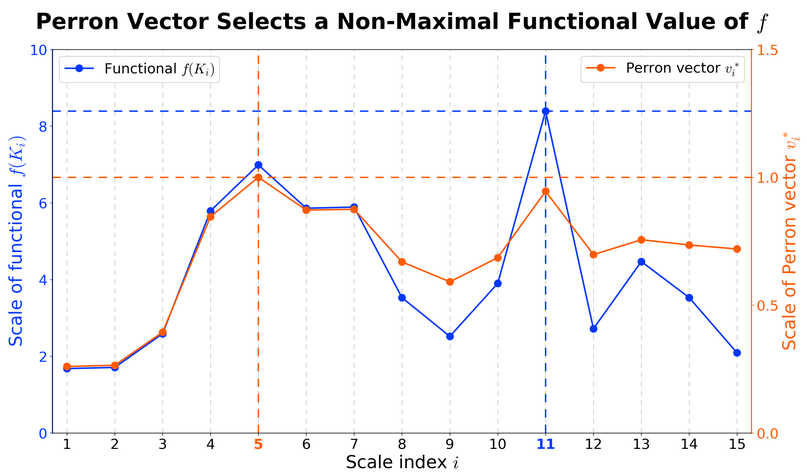
- Along the filtration, nonnegativity of f implies subadditivity.
- Σ_N is irreducible if $f(K_1) > 0$.
- By **Perron-Frobenius**, the principal eigenvector v^* is positive and unique up to scaling; it ranks the scales ϵ_i through their coupling in Σ_N.

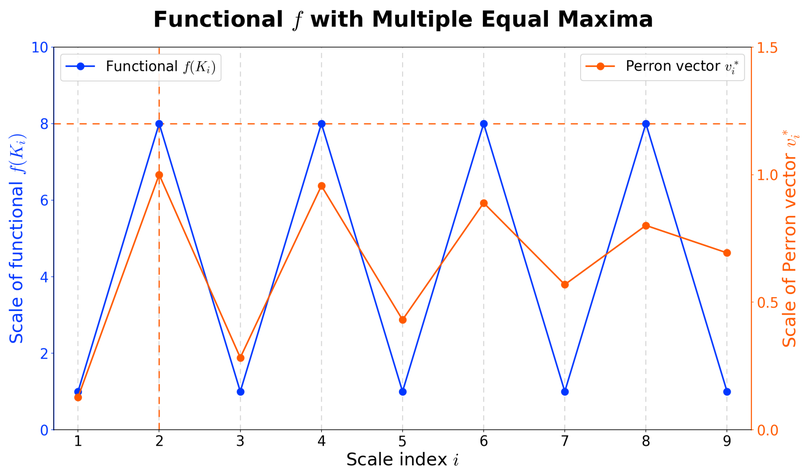
3. Non-monotone Scale Selection

The critical scale is chosen by the rule

$$i_{\text{crit}} \in \arg\max_i v_i^*, \quad \epsilon_{\text{crit}} = \epsilon_{i_{\text{crit}}}.$$

- $f(K_i)$ and v_i^* have the same local monotonicity pattern: $\operatorname{sgn}(v_{i+1}^* - v_i^*) = \operatorname{sgn}(f(K_{i+1}) - f(K_i))$.
- Thus, their **local maxima coincide**.
- For non-monotone f, global maxima need not coincide and the Perron vector can serve as a **tiebreaker** between separated equal maxima.
- For strictly monotone f, the Perron vector reflects the monotonicity and the critical scale lies at an endpoint.





4. Connection to Markov Chains

- Associate a **finite Markov chain** with the scale similarity matrix Σ_N.
- Construct its transition matrix P using **Doob's h-Transform**.
- Its stationary distribution is proportional to the squared coordinates of v^*:

$$P = \frac{1}{\lambda^*} D_{v^*}^{-1} \Sigma_N D_{v^*}, \qquad \pi_i^P = \frac{(v_i^*)^2}{\|v^*\|_2^2}.$$

Here $D_{v^*} = \operatorname{diag}(v_1^*, \dots, v_N^*)$. The matrix P maximizes the **Donsker-Varadhan free-energy functional** over irreducible, row-stochastic matrices:

$$\mathcal{F}(Q;\Sigma_N) \coloneqq \sum_{i,j=1}^{N} \pi_i^Q q_{ij} \log \frac{(\Sigma_N)_{ij}}{q_{ij}}, \qquad P = \arg\max \Big\{\mathcal{F}(Q;\Sigma_N) : \underset{\text{row-stochastic and irreducible}}{Q\in\mathbb{R}^{N\times N}}\Big\}.$$

Thus, P is the **variationally optimal stochastic normalization** of Σ_N.

5. Monotone Scale Selection

For a strictly increasing functional, $0 < f(K_1) < \cdots < f(K_N)$:

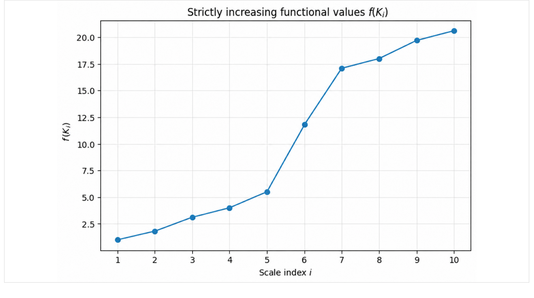
- Σ_N is positive definite and Σ_N^{-1} is tridiagonal.
- The **discrete Laplacian** $\mathcal{L} = f(K_1)^{-1}\operatorname{diag}(1,0,\dots,0) - \Sigma_N^{-1}$ is the generator of a continuous-time birth-death chain on $\{1,\dots,N\}$.

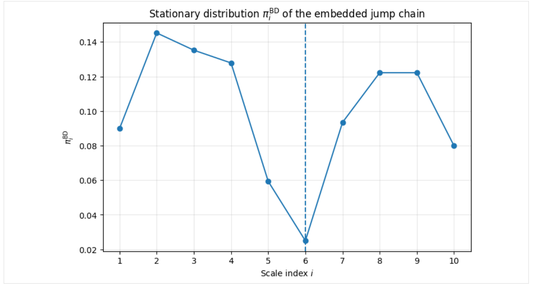
$$\mathcal{L} = \begin{pmatrix} -\frac{1}{d_2} & \frac{1}{d_2} & 0 & \cdots & 0 & 0 \\ \frac{1}{d_2} & -\left(\frac{1}{d_2}+\frac{1}{d_3}\right) & \frac{1}{d_3} & \ddots & \vdots & \vdots \\ 0 & \frac{1}{d_3} & -\left(\frac{1}{d_3}+\frac{1}{d_4}\right) & \ddots & 0 & 0 \\ \vdots & \ddots & \ddots & \ddots & \frac{1}{d_{N-1}} & 0 \\ 0 & \cdots & 0 & \frac{1}{d_{N-1}} & -\left(\frac{1}{d_{N-1}}+\frac{1}{d_N}\right) & \frac{1}{d_N} \\ 0 & \cdots & 0 & 0 & \frac{1}{d_N} & -\frac{1}{d_N} \end{pmatrix}, \qquad d_i = f(K_i) - f(K_{i-1}),\ i = 2,\dots,N.$$

For the embedded discrete-time chain, $P_{ij}^{\mathrm{BD}} = -(1-\delta_{ij})\mathcal{L}_{ij}/\mathcal{L}_{ii}$. Its stationary distribution gives the scale selection rule

$$i_{\text{crit}} \in \arg\min_i \pi_i^{\mathrm{BD}}, \qquad \pi_i^{\mathrm{BD}} = \frac{\mathcal{L}_{ii}}{\sum_{j=1}^{N} \mathcal{L}_{jj}}.$$

- For $2 \le i \le N-1$, $-\mathcal{L}_{ii} = d_i^{-1} + d_{i+1}^{-1}$.
- Minimizing π_i^{BD} favors two large consecutive increments.





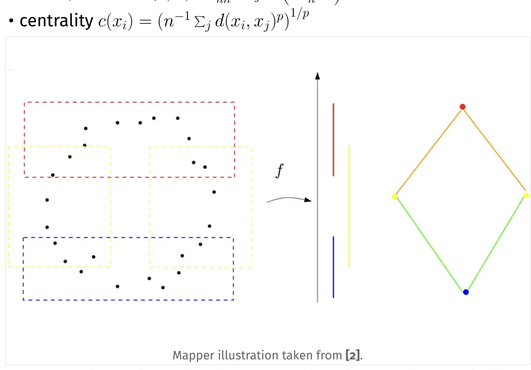
6. The Mapper Algorithm vs. Multiscale α-neighboring (MαN) approach

Let $X = \{x_1, \dots, x_n\} \subseteq \mathbb{R}^D$ be equipped with a metric d:

- choose a **lens function** $\ell : X \to \mathbb{R}^m$, typically $m \in \{1, 2\}$;
- choose a **finite open cover** $\mathcal{U} = \{U_i\}_{i\in I}$ of $\ell(X)$;
- form the pullback cover $X_i = \ell^{-1}(U_i)$.

Popular lens choices

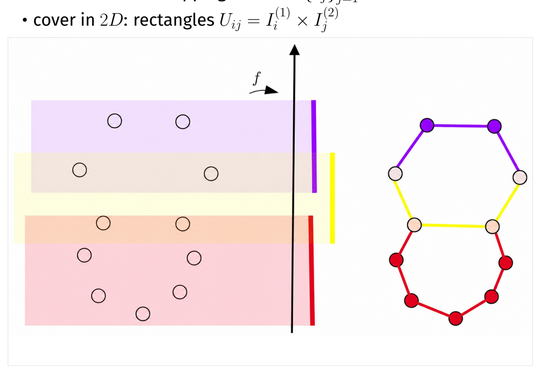
- density estimate $\rho(x_i) = \frac{1}{nh^D}\,\Sigma_j\, K\big(\frac{d(x_i,x_j)}{h}\big)$
- centrality $c(x_i) = (n^{-1}\,\Sigma_j\, d(x_i,x_j)^p)^{1/p}$



Mapper illustration taken from **[2]**.

Popular cover choices

- cover in $1D$: overlapping intervals $\{I_j\}_{j=1}^{N}$
- cover in $2D$: rectangles $U_{ij} = I_i^{(1)} \times I_j^{(2)}$



- Mapper **depends on the clustering algorithm** used on each X_i.
- Each cluster becomes a node; overlapping clusters are connected by an edge.
- MαN retains each subject as a node and removes clustering from graph construction; community detection may still be applied afterward.

The question becomes: by which rule should two nodes be connected?

7. MαN graph approach: Synthetic dataset experiment

Instead of clustering each X_i, connect $x, y \in X$ that belong to a common pullback set if and only if

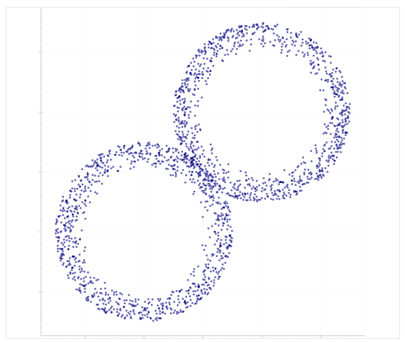
$$d(x,y) \le \max\{2\epsilon_i : i \in I, x, y \in X_i\}$$

Choose ϵ_i by applying spectral scale selection to X_i with

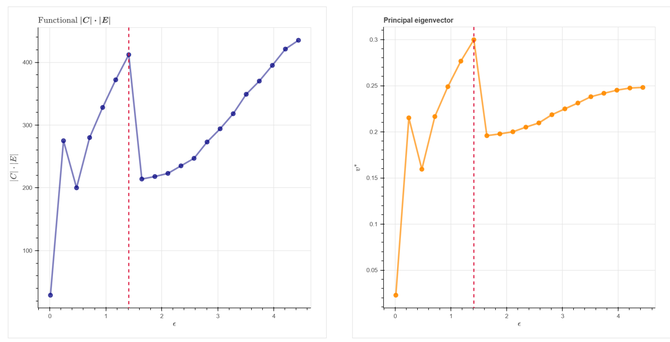
$$f(K) = C(K) \cdot E(K),$$

- $C(K)$ is the number of connected components and $E(K)$ the number of edges.
- Components merge while new edges appear, so the functional may exhibit **non-monotone** behavior.
- The graph **adapts its scale locally** instead of imposing one global threshold.
- First stress test: the **"noisy 8"** dataset.

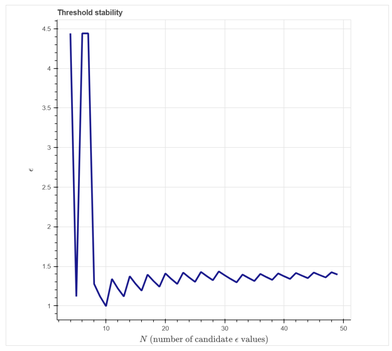
Noisy input X



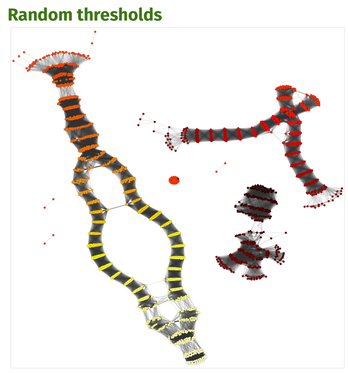
Select the local scales ϵ_i



Grid stability



Random thresholds



Spectral thresholds



8. Case Study: COVID-19 Spread in the USA

- Glushakov et al. **[3]** constructed a county-level graph to identify similarities in early COVID-19 trajectories, revealing four major communities associated with demographic and healthcare variables.
- Here, spectral scale selection determines the local MαN thresholds automatically using $f(K) = C(K)E(K)$.

County representation

- each node $x_i \in \mathbb{R}^{28}$ represents one of $3{,}142$ US counties and county equivalents
- first 14 entries: 13 normalized three-day case aggregates and their log-total
- second 14 entries: analogous death features

Lens and cover

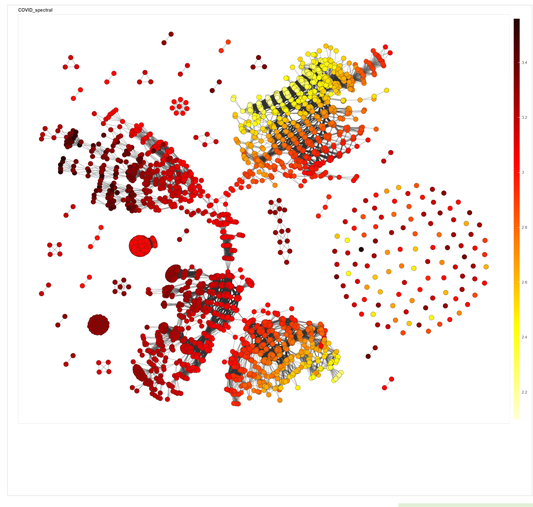
- chosen lens: $\ell(x) = (\rho(x), c(x))$
- chosen cover: overlapping rectangles $U_{ij} = I_i^{(1)} \times I_j^{(2)}$

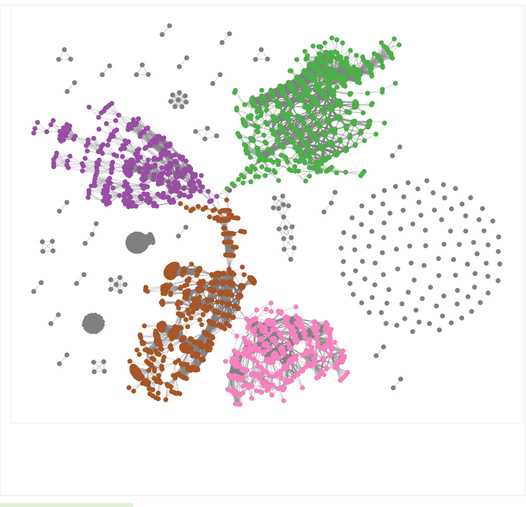
Similarity metric

Block-wise Euclidean metric on $\mathbb{R}^{14} \times \mathbb{R}^{14}$:

$$d(x_i, x_j) = \|x_i[1{:}14] - x_j[1{:}14]\|_2 + \|x_i[15{:}28] - x_j[15{:}28]\|_2$$

- The **Girvan–Newman algorithm [4]** yields four communities.
- **Left:** color indicates the centrality projection $c(x)$; darker colors represent higher values.
- **Right:** four colors indicate the four communities.





Four major communities identified by Girvan–Newman

Statistically significant differences in several county-level predictors not used to build the graph

Spectrally selected thresholds without manual tuning

References

[1] M. Schlather. *An Algebraic Generalization of the Entropy and Its Application to Statistics.* arXiv:2404.05854, 2024.

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